

Sales Performance Dashboard

End-to-End Business Intelligence Project — Executive Report

Prepared by: [Your Name]

Data Period: 2015 – 2018 | Superstore Sales Dataset

Executive Summary

This report documents an end-to-end analysis of four years of transactional sales data (9,994 order lines, 5,009 orders, \$2,297,200.65 in total revenue) to identify the drivers of profit performance across products, regions, and customer segments. While total revenue grew 51.4% between 2015 and 2018 and overall profit margin held steady near 12.5%, that stability masks significant localized problems: Furniture generates 32.3% of revenue but only 6.4% of profit, five states account for roughly \$78,000 in combined losses, and discounting above ~20% turns average transactions unprofitable. This report documents the full methodology — data cleaning, exploratory analysis, SQL modeling, and dashboard design — behind five targeted recommendations to recover profit without sacrificing revenue growth.

Objectives

- Quantify revenue, profit, and sales trends over the 2015–2018 period
- Identify which products, categories, and sub-categories help or hurt profitability
- Compare regional and state-level performance on margin, not just volume
- Segment customers by value, frequency, and profitability (RFM, ABC, CLV)
- Deliver specific, data-backed recommendations to improve profit

Dataset

Superstore Sales Dataset (Kaggle) — order-level retail transactions including order and ship dates, customer segment, geography (region/state/city), product category/sub-category, sales, quantity, discount, and profit. After cleaning, the analysis dataset contains 9,994 valid order lines across 34 columns (12 engineered on top of the original 22).

Order ID	Order Date	Customer Name	Segment	Region	Category	Sub-Category	Sales	Discount	Profit
CA-2017-152156	2017-11-08	Claire Gute	Consumer	South	Furniture	Bookcases	261.96	0.0	41.91
CA-2017-152156	2017-11-08	Claire Gute	Consumer	South	Furniture	Chairs	731.94	0.0	219.58
CA-2017-138688	2017-06-12	Darrin Van Huff	Corporate	West	Office Supplies	Labels	14.62	0.0	6.87
US-2016-108966	2016-10-11	Sean O'Donnell	Consumer	South	Furniture	Tables	957.58	0.45	-383.03
US-2016-108966	2016-10-11	Sean O'Donnell	Consumer	South	Office Supplies	Storage	22.37	0.2	2.52
CA-2015-115812	2015-06-09	Brosina Hoffman	Consumer	West	Furniture	Furnishings	48.86	0.0	14.17
CA-2015-115812	2015-06-09	Brosina Hoffman	Consumer	West	Office Supplies	Art	7.28	0.0	1.97
CA-2015-115812	2015-06-09	Brosina Hoffman	Consumer	West	Technology	Phones	907.15	0.2	90.72
CA-2015-115812	2015-06-09	Brosina Hoffman	Consumer	West	Office Supplies	Binders	18.5	0.2	5.78
CA-2015-115812	2015-06-09	Brosina Hoffman	Consumer	West	Office Supplies	Appliances	114.9	0.0	34.47

Figure 1: Sample records from the cleaned dataset.

Methodology

The raw export was cleaned in Python (Pandas/NumPy) using a modular, logged, and validated pipeline. A structural issue was diagnosed where three separate source sheets (Orders, People, Returns) had been concatenated into one file — this was identified and corrected by isolating genuine Orders rows via the Row ID column, rather than treating the resulting nulls as ordinary missing data. 504 duplicate rows were removed, business-rule filters were applied (positive Sales/Quantity, Discount in [0,1], Ship Date ≥ Order Date), and 12 features were engineered, including Profit Margin, calendar fields, Sales/Discount tiers, and Customer Type.

```

1 # ---
2 # jupyter:
3 #   jupyter:
4 #     text_representation:
5 #       extension: .py
6 #       format_name: percent
7 #     kernelspec:
8 #       display_name: Python 3
9 #       language: python
10 #       name: python3
11 # ---
12
13 # %% [markdown]
14 # # Sales Performance Dashboard – Data Cleaning & Feature Engineering
15 #
16 # **Dataset:** Superstore Sales Dataset (Kaggle)
17 # **Purpose:** Clean raw transactional sales data and engineer the features used
18 # in EDA, SQL analysis, and the Power BI dashboard.
19 #
20 # This notebook is saved in Jupyter "percent" format – open it directly in
21 # ...
22
23 # %% [markdown]
24 # ## 3. Diagnose the structural data-quality issue
25 #
26 # This raw export flattens three separate Excel sheets (Orders, People,
27 # Returns) into one CSV. Rows belonging to the People>Returns sheets show up
28 # as near-total nulls across every Orders column – they aren't missing data,
29 # they simply don't belong in this table. `Row ID` is a clean sequential
30 # integer that only the real Orders sheet has, so it's used as the structural
31 # filter, rather than trying to impute values into rows that were never
32 # Orders rows in the first place.
33
34 # %%
35 def isolate_orders_table(df: pd.DataFrame) -> pd.DataFrame:
36     before = len(df)
37     is_orders_row = pd.to_numeric(df["Row ID"], errors="coerce").notnull()
38     df = df.loc[is_orders_row].copy()
39     df["Row ID"] = df["Row ID"].astype(int)
40     log.info(
41         "Isolated Orders rows: kept %s of %s (%s non-Orders rows dropped)",
42         len(df), before, before - len(df),
43     )
44     return df
45
46 # %% [markdown]

```

Figure 2: Excerpt from the data cleaning notebook (01_data_cleaning.py).

Exploratory Data Analysis

20 visualizations were produced covering sales and profit distributions, category/sub-category performance, regional and state comparisons, top and bottom products, discount-tier impact on margin, shipping-mode performance, monthly/quarterly/yearly trends, a correlation matrix and pairplot across key numeric fields, and RFM / ABC (Pareto) / Customer-Lifetime-Value segmentation. Every chart is paired with a printed Business Insight and Business Recommendation in the notebook. Selected charts are shown below; the full set is included in the accompanying code package.

```

1 # ---
2 # jupyter:
3 #   jupyter:
4 #     text_representation:
5 #       extension: .py
6 #       format_name: percent
7 #     kernelspec:
8 #       display_name: Python 3
9 #       language: python
10 #       name: python3
11 # ---
12
13 # %% [markdown]
14 # Sales Performance Dashboard — Exploratory Data Analysis (EDA)
15 #
16 # Reads the cleaned dataset and produces 20 charts (saved to `./images/`)
17 # plus a printed Business Insight + Business Recommendation after
18 # every visualization. Every number quoted in the business insights report
19 # and the executive PDF report traces back to this notebook.
20 #
21 # Covers: sales/profit distributions, revenue & profit trend, category &
22 # sub-category analysis, region & state analysis, top & bottom
23 # products/customers, discount analysis, shipping analysis, monthly/
24 # quarterly/yearly trends, boxplots, correlation matrix, pairplot, and
25 # RFM / ABC (Pareto) / Customer-Lifetime-Value segmentation.
26
27
28 ...
29
30
31
32
33 # %% [markdown]
34 # ## Chart 1 — Sales Distribution
35 #
36 # %%
37 plt.figure(figsize=(8, 5))
38 sns.histplot(df["Sales"], bins=60, kde=True, color="#2563eb")
39 plt.title("Sales Distribution per Order Line", fontsize=13, fontweight="bold")
40 plt.xlabel("Sales ($)"); plt.ylabel("Number of Order Lines")
41 plt.xlim(0, 2000)
42 savefig("01_sales_distribution")
43 report(1, "Sales Distribution",
44        f"Median order-line sales is ${df['Sales'].median():.2f} vs. a mean of ${df['Sales'].mean():.2f} "
45        "— a right-skewed distribution driven by a small number of large orders.",
46        "Don't use the mean alone for pricing/target-setting conversations; report median alongside mean "
47        "to avoid overstating what a 'typical' order looks like.")
48
49
50 # %% [markdown]
51 # ## Chart 2 — Profit Distribution
52 #
53 # %%
54 plt.figure(figsize=(8, 5))
55 sns.histplot(df["Profit"], bins=80, kde=True, color="#16a34a")
56 plt.title("Profit Distribution per Order Line", fontsize=13, fontweight="bold")
57 plt.xlabel("Profit ($)"); plt.ylabel("Number of Order Lines")
58 plt.xlim(-500, 500)
59 savefig("02_profit_distribution")
60 pct_unprofitable = (df["Profit"] < 0).mean() * 100
61 report(2, "Profit Distribution",
62        f"{pct_unprofitable:.1f}% of all order lines are sold at a loss.",
63        "Flag loss-making order lines at the point of sale (e.g. a discount-approval workflow) rather than "
64        "discovering them only in quarterly reporting.")

```

Figure 3: Excerpt from the EDA notebook (02_eda.py).

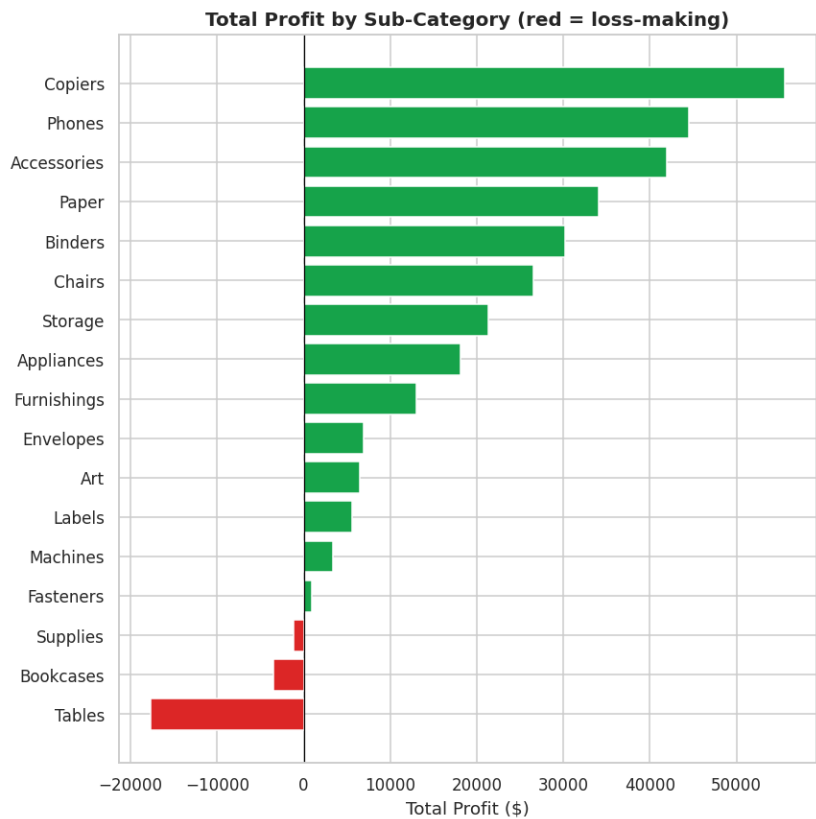


Figure 4: Total Profit by Sub-Category — Tables, Bookcases, and Supplies are loss-making.

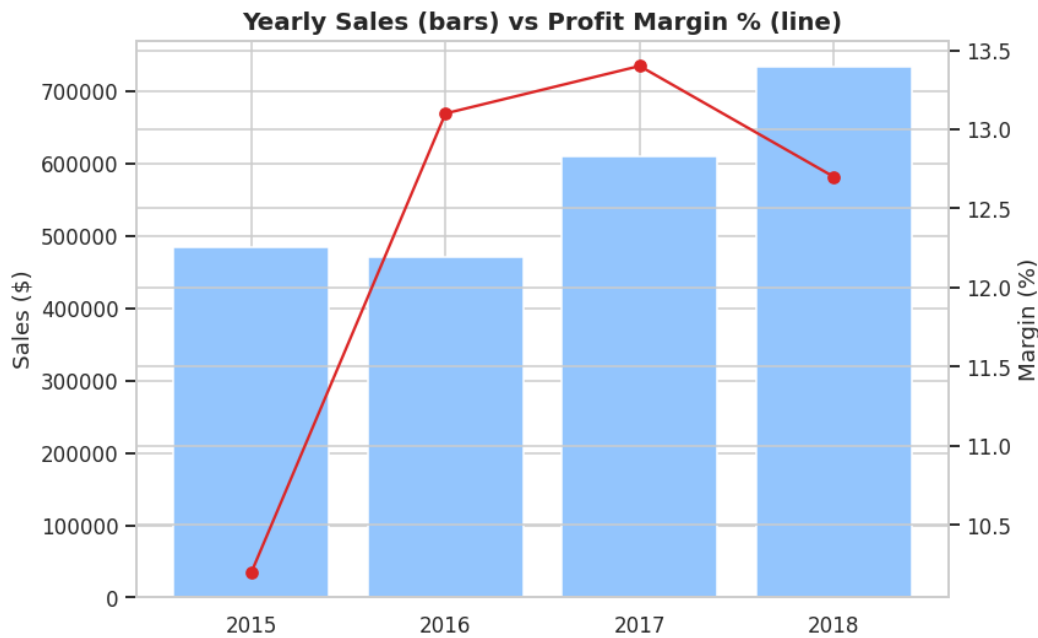


Figure 5: Yearly Sales (bars) vs Profit Margin % (line), 2015–2018.

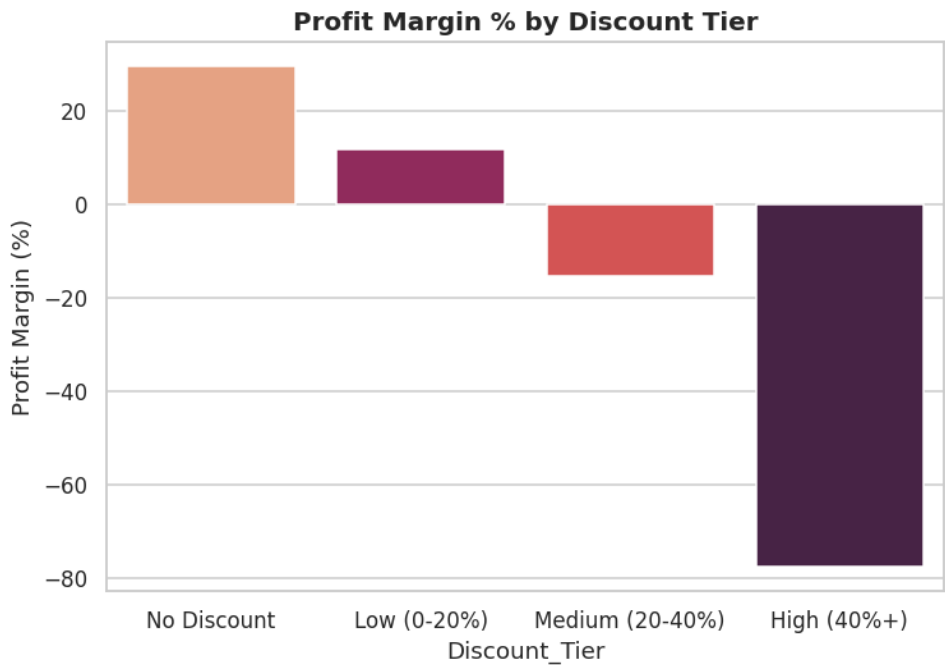


Figure 6: Profit Margin collapses as Discount Tier increases.

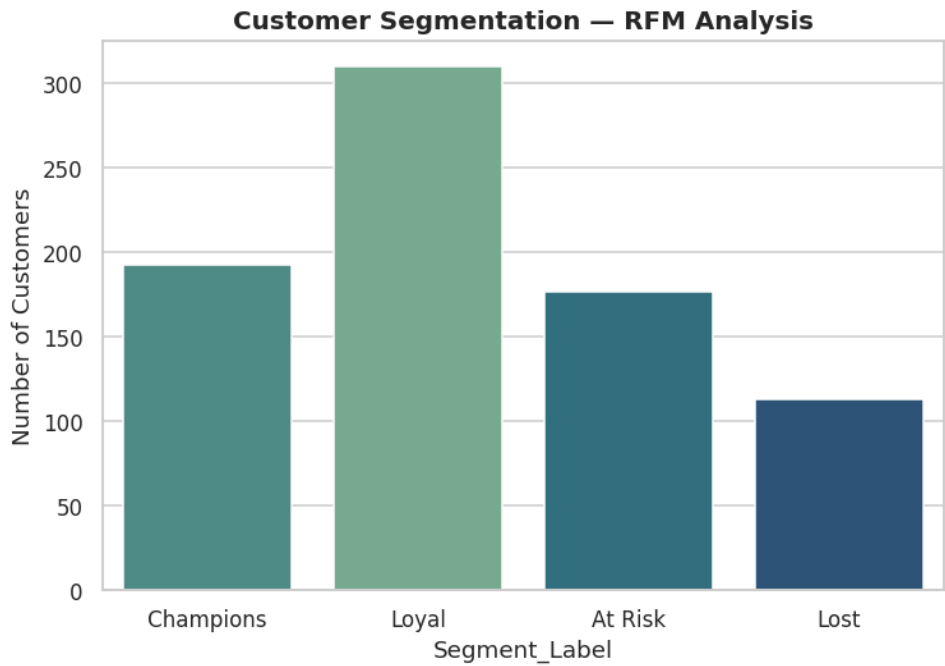


Figure 7: RFM Customer Segmentation — Champions drive a disproportionate share of revenue.

SQL Analysis

A normalized star schema (fact_orders + dim_customers, dim_products, dim_locations) was built in SQLite and queried with 40 business questions covering top/worst products, regional and state profitability, year-over-year/quarter-over-quarter/month-over-month growth, running and rolling totals, moving averages, Pareto/NTILE customer segmentation, repeat-purchase analysis, and reusable views. Every query was executed and verified against the live database — for example, the top 20% of customers by revenue were confirmed to generate 48.15% of total sales.

```

1  -- =====
2  -- Sales Performance Dashboard - Database Schema (Star Schema)
3  -- =====
4  -- Design note: the raw Superstore export is a single flat table. It was
5  -- deliberately normalized into a star schema here (one fact table +
6  -- three dimension tables) to demonstrate real relational design and to
7  -- make JOIN-based queries meaningful rather than cosmetic.
8  -- =====
9
10 CREATE TABLE dim_customers (
11     customer_id    TEXT PRIMARY KEY,
12     customer_name  TEXT NOT NULL,
13     segment        TEXT NOT NULL          -- Consumer / Corporate / Home Office
14 );
15
16 CREATE TABLE dim_products (
17     product_id     TEXT PRIMARY KEY,
18     product_name   TEXT NOT NULL,
19     category       TEXT NOT NULL,        -- Furniture / Office Supplies / Technology
20     sub_category   TEXT NOT NULL
21 );
22
23 CREATE TABLE dim_locations (
24     location_id    INTEGER PRIMARY KEY,
25     city           TEXT,
26     state          TEXT,
27     region         TEXT,                -- East / West / Central / South
28     postal_code    INTEGER,
29     country        TEXT
30 );
31
32 CREATE TABLE fact_orders (
33     row_id         INTEGER PRIMARY KEY,
34     order_id       TEXT NOT NULL,
35     order_date     DATE NOT NULL,
36     ship_date      DATE NOT NULL,
37     ship_mode      TEXT,
38     customer_id    TEXT REFERENCES dim_customers(customer_id),
39     product_id     TEXT REFERENCES dim_products(product_id),
40     location_id    INTEGER REFERENCES dim_locations(location_id),
41     sales          REAL NOT NULL,
42     quantity       INTEGER NOT NULL,
43     discount       REAL NOT NULL,
44     profit         REAL NOT NULL,
45     profit_margin  REAL,

```

Figure 8: Star schema DDL (00_schema.sql).

```

1  -- =====
2  -- Sales Performance Dashboard - 30 Business SQL Queries
3  -- Dialect: SQLite (all queries tested against sql/superstore.db)
4  -- =====
5  -- Each query is preceded by: the business question it answers, and a
6  -- one-line explanation of the technique used.
7  -- =====
8
9
10     END AS profit_tier,
11     COUNT(*) AS num_orders,
12     ROUND(SUM(sales), 2) AS total_sales
13 FROM fact_orders
14 GROUP BY profit_tier
15 ORDER BY total_sales DESC;
16
17
18 -- Q11. Monthly revenue trend (year + month)
19 -- Technique: multi-column GROUP BY, ORDER BY chronological
20 SELECT order_year, order_month, ROUND(SUM(sales), 2) AS monthly_revenue
21 FROM fact_orders
22 GROUP BY order_year, order_month
23 ORDER BY order_year, order_month;
24
25
26 -- Q12. Year-over-year revenue growth %
27 -- Technique: CTE + window function LAG()
28 WITH yearly AS (
29     SELECT order_year, SUM(sales) AS total_sales
30     FROM fact_orders
31     GROUP BY order_year
32 )
33 SELECT order_year,
34        ROUND(total_sales, 2) AS total_sales,
35        ROUND(total_sales - LAG(total_sales) OVER (ORDER BY order_year), 2) AS yoy_change,
36        ROUND((total_sales - LAG(total_sales) OVER (ORDER BY order_year)) * 100.0
37              / LAG(total_sales) OVER (ORDER BY order_year), 2) AS yoy_growth_pct
38 FROM yearly
39 ORDER BY order_year;

```

Figure 9: Sample business queries (01_business_queries.sql).

Dashboard

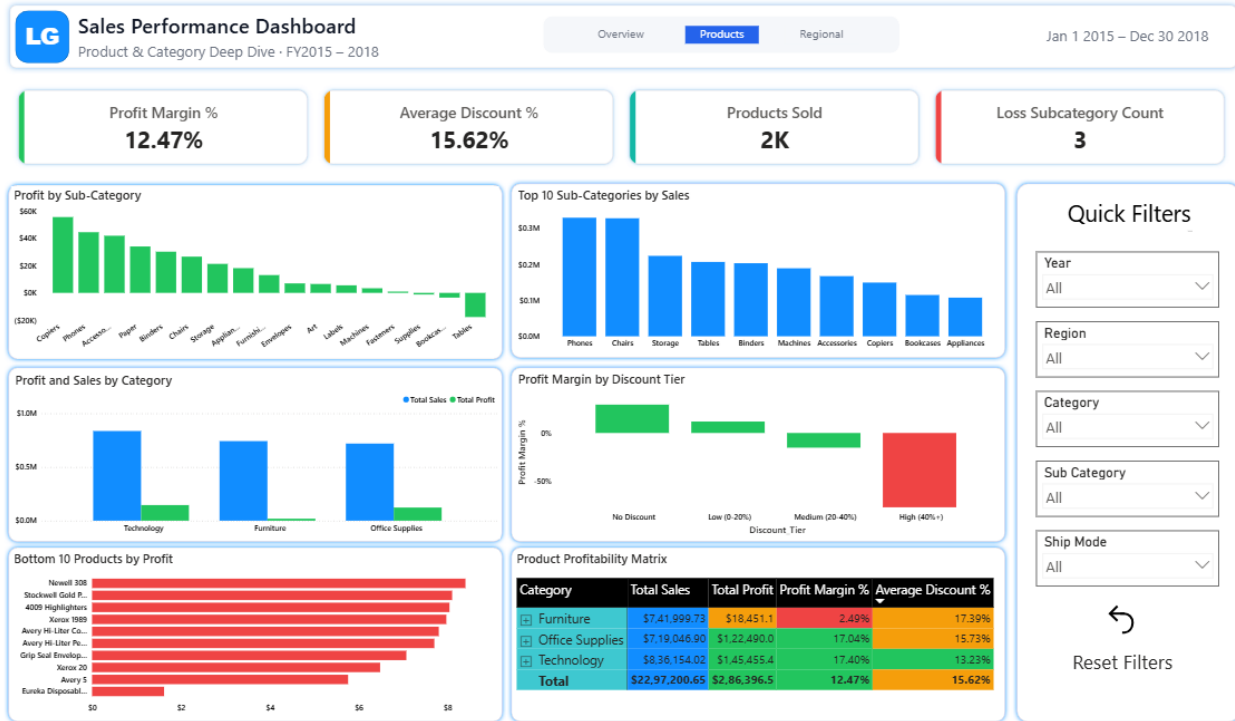
A 3-page Power BI dashboard was designed and built: an Executive Overview (KPI cards, monthly trend, category and regional mix, top products/customers), a Product & Category Deep Dive (sub-category profit ranking, discount-tier margin analysis, a category profitability matrix), and a Regional & Customer Performance page (state-level profit analysis, RFM/segment mix, a Top-20-customers matrix). All pages share a consistent color system — blue for sales, green for profit, red for loss, amber for caution — applied via conditional formatting rules throughout, not static color choices.



Source: Superstore sales dataset (2015–2018)

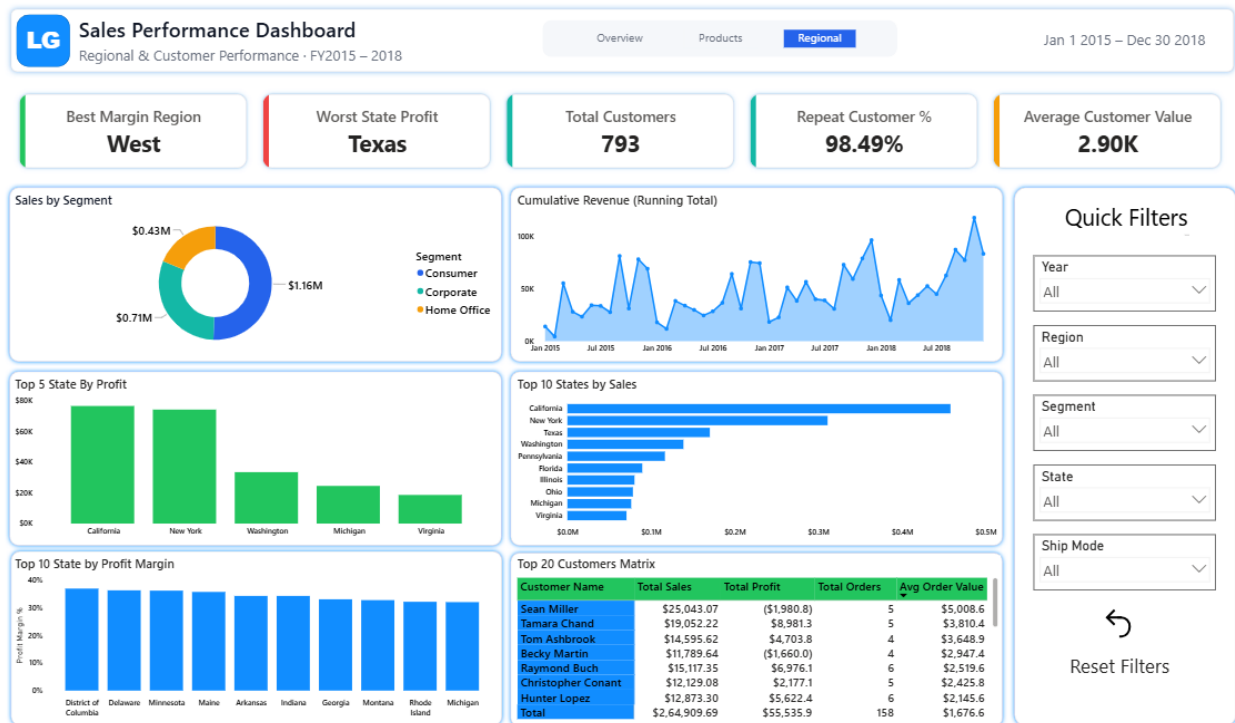
Page 1 of 3

Figure 10: Dashboard Page 1 — Executive Overview.



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Figure 11: Dashboard Page 2 — Product & Category Deep Dive.



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Figure 12: Dashboard Page 3 — Regional & Customer Performance.

Business Insights

Metric	Finding
Revenue growth (2015–18)	+51.4% (\$484K → \$733K)
Overall profit margin	12.47% (improved +2.5pp over the period)
Furniture category	32.3% of revenue, only 6.4% of profit (2.5% margin)
Worst sub-category	Tables (-8.6% margin, -\$17,726 total loss)
Worst state	Texas (-\$25,729 total profit)
Best region	West (14.9% margin) vs Central (7.9%)
Discount impact	Margin falls from 29.5% (no discount) to -77.4% (40%+ discount)
RFM "Champions" segment	24% of customers drive ~40% of revenue
Repeat customer revenue share	99.8% of total revenue
Top 20% of customers	Generate 48.15% of total revenue (SQL-verified)

Full list of 21 insights, each grounded in the EDA notebook's computed output, is included in the project's `business_insights.md` file within the code package.

Recommendations

1. Cap discounts on Furniture — especially Tables — at 20%. Discounting past this threshold turns a marginal product deeply unprofitable; Furniture's discount-profit correlation (-0.48) is more than double the company-wide figure (-0.22), confirming discounting practice, not underlying cost, is the driver.
2. Open a targeted profitability review in Texas, Ohio, and Pennsylvania. These three states alone account for roughly \$58,000 in lost profit — investigate whether this is a discounting pattern, a shipping-cost issue, or a sales-incentive structure specific to these markets.
3. Prioritize marketing and inventory investment in Technology and the West region. Both show the strongest margin profile in the business and the best expected return on incremental investment.
4. Launch a retention/loyalty program for the RFM "Champions" segment rather than broad customer-acquisition spend. With 98.5% of customers already repeat buyers and Champions alone driving ~40% of revenue, protecting this specific group outperforms undifferentiated marketing.
5. Route the confirmed high-volume, negative-profit product list through a repricing or supplier-cost renegotiation process. These products are quantified, ongoing losses at meaningful scale, not marginal or ambiguous cases.

Conclusion

The evidence does not support broad pricing or marketing changes — the business is fundamentally healthy, with revenue growth and stable overall margin. The opportunity is precise: a small number of specific sub-categories, states, and discounting practices are responsible for a disproportionate share of lost profit, and each has a clear, low-risk, testable fix. Addressing these five items is the fastest, lowest-risk path to measurable margin improvement without putting current revenue growth at risk.